Qubit Initialization in Variational Quantum Portfolio Optimization

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Selecting asset allocations that jointly optimize return and control downside risk remains a central challenge in com

variational quantum algorithms

Introduction Constructing portfolios that balance expected return against financial risk remains a core optimizatio

where $\mathbf{w}$ is the portfolio weight vector and $q \in [0, 1]$ is the risk parameter. For a binary asset selection problem with n a

Quantum computing addresses this combinatorial structure directly by encoding all 2^n asset combinations as quant

A critical but underexplored aspect of VQAs is the choice of the initial quantum state. Two common strategies are

In this work, we investigate the impact of qubit initialization strategy on variational quantum portfolio optimizatio

The remainder of this paper is as follows. Section 2 reviews related work. Section 3 introduces background concept

Related Work

Portfolio optimization has been studied extensively since Markowitz [?] introduced the mean-variance framework, w

Within the quantum computing literature, two principal variational strategies have emerged for combinatorial optir

Barkoutsos et al. [?] demonstrated that replacing the standard expectation value with CVaR as the objective funct

A related but distinct line of investigation concerns parameter and state initialization in variational circuits. McCle

Background Concepts

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